

Cape Town, Western Cape 7441 • jaswinchinthala08@gmail.com • +27679311089

[Jaswin Chinthala | LinkedIn](#) • [jaswin-codes \(Jaswin Chinthala\)](#) • [Jaswin Chinthala Portfolio](#)

Education

University of Cape Town (UCT)

Cape Town, South Africa

BSc Eng in Electrical Engineering, Mechatronics Focus

Expected Graduation: November 2029

- Relevant Coursework: Computer Science I (Python & Algorithms), Introduction to Electronic Engineering, Engineering Calculus (Differential & Integral Calculus), Engineering Physics (Mechanics & Thermodynamics)
- Elected 1 of 6 class representatives for a 170-student cohort; relayed student concerns to the TA, driving timely resolution and communication.

Blouberg International School

Cape Town, South Africa

Cambridge International A-Levels & AS-Levels

November 2025

- A-Levels: Mathematics (A), Physics (B)
- AS-Levels: Business (A), Computer Science (B), Chemistry (B), English (B)

Experience

EMi Lab (UCT)

Cape Town, South Africa

Vacation Work Student (Electric Mobility Project)

June 2026 – July 2026

- Built foundational skills in MATLAB coursework covering Battery Management system modeling, cell balancing, and state-of-health tracking via Simulink and Simscape simulations in a team of 5 members.
- Authored a technical report on swappable battery architectures through academic literature reviews to propose 1 infrastructure configuration out of 4 for modular electric tuk-tuks.

UCT Racing (Formula Student Team)

Cape Town, South Africa

EV Research Subteam Lead & Power Subsystem Member

February 2026 – Present

- Incoming Subteam Lead for newly formed EV Research subteam; tasked with building the subteam's research focus on electric drivetrain architectures and battery chemistry beginning August 2026.
- Co-authored technical documentation for cooling architecture, specifically X2.4 fan integration.
- Researching telemetry systems for a technical report on real-time vehicle data communication while building knowledge of battery thermal management.

Scientific Caribbean Foundation and UCT

Cape Town, South Africa

Undergraduate Student Researcher (STEM Methodology Workshop)

May 2026

- Formulated an independent, simulation-based research framework using Python for post-harvest produce deterioration, adapting core concepts from the University of Florida's FreshID project to low-cost constraints.
- Evaluated simulated hyperspectral signatures and biochemical markers to track produce freshness, presenting empirical findings (successfully demonstrating that specific spectral regions like Red-edge and NIR are highly predictive of freshness) via a technical poster to an academic panel

Technical Projects

AfriGuard: Multilingual Safety Red-Teaming Framework (Team of 4)

Cape Town, South Africa

Global South AI Safety Hackathon (Africa Track)

June 2026

- Engineered an automated Python-based multilingual red-teaming pipeline for 4 frontier LLMs through 1,120 adversarial runs, exposing a 50.1% mean Attack Success Rate in African languages visualized on a Streamlit dashboard.

Skeletal Interface and Gesture Network (Naruto Jutsu)

Side Project

March 2026

- Built a real-time hand-gesture recognition system using OpenCV and 21-point landmark tracking to trigger automation scripts without custom model training

Skills & Interests

Technical: Python (OpenCV, MediaPipe, Selenium), C Programming, MATLAB, Simulink, Simscape and PyAutoGUI.

Laboratory: Arduino Microcontroller Prototyping, mmWave Radar Sensor Exposure

Interests: Chess, Hiking & Long-Distance Running, Mangas & Books